Exercise Sheet 5 - Solutions
Propositional Logic – Semantics & Equivalences

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1. For example:
   - a non-valid satisfiable formula: $(p \to p) \wedge (\neg q \vee r)$. It is satisfiable because $p = \mathbf{T}, q = \mathbf{T}, r = \mathbf{T}$ for example satisfies, while $p = \mathbf{T}, q = \mathbf{T}, r = \mathbf{F}$ does not.
   - a non-unsatisfiable, falsifiable formula: $(p \to p) \wedge (\neg q \vee r)$ (same as above). It is falsifiable because $p = \mathbf{T}, q = \mathbf{T}, r = \mathbf{F}$ for example falsifies it, while $p = \mathbf{T}, q = \mathbf{T}, r = \mathbf{T}$ does not (it satisfies it).
   - an unsatisfiable formula: $\neg((p \wedge q) \to (p \vee q))$
   - a valid formula: $(p \wedge q) \to (\neg(\neg p \vee \neg q))$

2. Here is the truth table for $F = (q \vee r \to p) \wedge (q \to r) \wedge \neg r$:

p	q	r	$q \vee r$	$q \vee r \to p$	$q \to r$	$\neg r$	F
T	**T**	**T**	**T**	**T**	**T**	**F**	**F**
T	**T**	**F**	**T**	**T**	**F**	**T**	**F**
T	**F**	**T**	**T**	**T**	**T**	**F**	**F**
T	**F**	**F**	**F**	**T**	**T**	**T**	**T**
F	**T**	**T**	**T**	**F**	**T**	**F**	**F**
F	**T**	**F**	**T**	**F**	**F**	**T**	**F**
F	**F**	**T**	**T**	**F**	**T**	**F**	**F**
F	**F**	**F**	**F**	**T**	**T**	**T**	**T**

   Because there are rows where F is **F**, the formula is not valid.

3. To convert $(q \vee r \to p) \wedge (q \to r) \wedge \neg r$ to a DNF, we first enumerate all **T** rows:
   - $p \wedge \neg q \wedge \neg r$
   - $\neg p \wedge \neg q \wedge \neg r$

   We then combine these formulas using $\vee$:

$$(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

4. To convert $(q \vee r \to p) \wedge (q \to r) \wedge \neg r$ to a CNF, we first enumerate all the **F** rows:
   - $p \wedge q \wedge r$
   - $p \wedge q \wedge \neg r$
   - $p \wedge \neg q \wedge r$
   - $\neg p \wedge q \wedge r$
   - $\neg p \wedge q \wedge \neg r$
   - $\neg p \wedge \neg q \wedge r$

We combine the negation of those using $\land$:

$$\neg(p \land q \land r) \land \neg(p \land q \land \neg r) \land \neg(p \land \neg q \land r) \land \neg(\neg p \land q \land r) \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$$

We then use de Morgan and double negation elimination to convert this formula to a CNF as follows:

- $\underline{\neg(p \land q \land r)} \land \neg(p \land q \land \neg r) \land \neg(p \land \neg q \land r) \land \neg(\neg p \land q \land r) \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land \underline{\neg(p \land q \land \neg r)} \land \neg(p \land \neg q \land r) \land \neg(\neg p \land q \land r) \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor \underline{\neg\neg r}) \land \neg(p \land \neg q \land r) \land \neg(\neg p \land q \land r) \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land \underline{\neg(p \land \neg q \land r)} \land \neg(\neg p \land q \land r) \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor \underline{\neg\neg q} \lor \neg r) \land \neg(\neg p \land q \land r) \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor q \lor \neg r) \land \underline{\neg(\neg p \land q \land r)} \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor q \lor \neg r) \land (\underline{\neg\neg p} \lor \neg q \lor \neg r) \land \neg(\neg p \land q \land \neg r) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor q \lor \neg r) \land (p \lor \neg q \lor \neg r) \land \underline{\neg(\neg p \land q \land \neg r)} \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor q \lor \neg r) \land (p \lor \neg q \lor \neg r) \land (\underline{\neg\neg p} \lor \neg q \lor \underline{\neg\neg r}) \land \neg(\neg p \land \neg q \land r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor q \lor \neg r) \land (p \lor \neg q \lor \neg r) \land (p \lor \neg q \lor r) \land \underline{\neg(\neg p \land \neg q \land r)}$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor q \lor \neg r) \land (p \lor \neg q \lor \neg r) \land (p \lor \neg q \lor r) \land (\underline{\neg\neg p} \lor \underline{\neg\neg q} \lor \neg r)$
- $(\neg p \lor \neg q \lor \neg r) \land (\neg p \lor \neg q \lor r) \land (\neg p \lor q \lor \neg r) \land (p \lor \neg q \lor \neg r) \land (p \lor \neg q \lor r) \land (p \lor q \lor \neg r)$

5. We now convert $(q \lor r \to p) \land (q \to r) \land \neg r$ to a CNF using equivalences

   - $(\underline{q \lor r \to p}) \land (q \to r) \land \neg r$
   - $(\neg(q \lor r) \lor p) \land (\underline{q \to r}) \land \neg r$ – implication elimination
   - $(\underline{\neg(q \lor r)} \lor p) \land (\neg q \lor r) \land \neg r$ – implication elimination
   - $(\underline{(\neg q \land \neg r) \lor p}) \land (\neg q \lor r) \land \neg r$ – de Morgan
   - $(\neg q \lor p) \land (\neg r \lor p) \land (\neg q \lor r) \land \neg r$ – distributivity of $\lor$ over $\land$

6. Here is the truth table for $F = (q \lor r \to p) \lor (q \to r) \lor \neg r$:

p	q	r	$q \lor r$	$q \lor r \to p$	$q \to r$	$\neg r$	F
T	**T**	**T**	**T**	**T**	**T**	**F**	**T**
T	**T**	**F**	**T**	**T**	**F**	**T**	**T**
T	**F**	**T**	**T**	**T**	**T**	**F**	**T**
T	**F**	**F**	**F**	**T**	**T**	**T**	**T**
F	**T**	**T**	**T**	**F**	**T**	**F**	**T**
F	**T**	**F**	**T**	**F**	**F**	**T**	**T**
F	**F**	**T**	**T**	**F**	**T**	**F**	**T**
F	**F**	**F**	**F**	**T**	**T**	**T**	**T**

   Because F is **T** w.r.t. all valuations, the formula is therefore valid.

7. Consider the following valuation ϕ: $\phi(p_0) = \mathbf{T}$, $\phi(p_1) = \mathbf{F}$, $\phi(q_0) = \mathbf{F}$, $\phi(q_1) = \mathbf{T}$, $\phi(r_0) = \mathbf{F}$, $\phi(r_1) = \mathbf{T}$, $\phi(s_0) = \mathbf{T}$, $\phi(s_1) = \mathbf{F}$. Therefore,

   - $\phi(p_0 \lor p_1) = \mathbf{T}$ because $\phi(p_0) = \mathbf{T}$
   - $\phi(q_0 \lor q_1) = \mathbf{T}$ because $\phi(q_1) = \mathbf{T}$
   - $\phi(r_0 \lor r_1) = \mathbf{T}$ because $\phi(r_1) = \mathbf{T}$

- $\phi(s_0 \vee s_1) = \mathbf{T}$ because $\phi(s_0) = \mathbf{T}$
- therefore $\phi((p_0 \vee p_1) \wedge (q_0 \vee q_1) \wedge (r_0 \vee r_1) \wedge (s_0 \vee s_1)) = \mathbf{T}$ because all conjuncts evaluate to $\mathbf{T}$
- $\phi(\neg p_0 \vee \neg p_1) = \mathbf{T}$ because $\phi(p_1) = \mathbf{F}$, and therefore $\phi(\neg p_1) = \mathbf{T}$
- $\phi(\neg q_0 \vee \neg q_1) = \mathbf{T}$ because $\phi(q_0) = \mathbf{F}$, and therefore $\phi(\neg q_0) = \mathbf{T}$
- $\phi(\neg r_0 \vee \neg r_1) = \mathbf{T}$ because $\phi(r_0) = \mathbf{F}$, and therefore $\phi(\neg r_0) = \mathbf{T}$
- $\phi(\neg s_0 \vee \neg s_1) = \mathbf{T}$ because $\phi(s_1) = \mathbf{F}$, and therefore $\phi(\neg s_1) = \mathbf{T}$
- therefore $\phi((\neg p_0 \vee \neg p_1) \wedge (\neg q_0 \vee \neg q_1) \wedge (\neg r_0 \vee \neg r_1) \wedge (\neg s_0 \vee \neg s_1)) = \mathbf{T}$ because all conjuncts evaluate to $\mathbf{T}$
- $\phi(p_0 \vee q_0) = \mathbf{T}$ because $\phi(p_0) = \mathbf{T}$
- $\phi(r_0 \vee s_0) = \mathbf{T}$ because $\phi(s_0) = \mathbf{T}$
- $\phi(p_1 \vee q_1) = \mathbf{T}$ because $\phi(q_1) = \mathbf{T}$
- $\phi(r_1 \vee s_1) = \mathbf{T}$ because $\phi(r_1) = \mathbf{T}$
- $\phi(p_0 \vee r_0) = \mathbf{T}$ because $\phi(p_0) = \mathbf{T}$
- $\phi(q_0 \vee s_0) = \mathbf{T}$ because $\phi(s_0) = \mathbf{T}$
- $\phi(p_1 \vee r_1) = \mathbf{T}$ because $\phi(r_1) = \mathbf{T}$
- $\phi(q_1 \vee s_1) = \mathbf{T}$ because $\phi(q_1) = \mathbf{T}$
- Therefore $\phi((p_0 \vee q_0) \wedge (r_0 \vee s_0) \wedge (p_1 \vee q_1) \wedge (r_1 \vee s_1) \wedge (p_0 \vee r_0) \wedge (q_0 \vee s_0) \wedge (p_1 \vee r_1) \wedge (q_1 \vee s_1)) = \mathbf{T}$ because all conjuncts evaluate to $\mathbf{T}$
- Finally the entire formula evaluate to $\mathbf{T}$ because all conjuncts evaluate to $\mathbf{T}$

The truth table would have $2^8 = 256$ rows because we have 8 atoms.